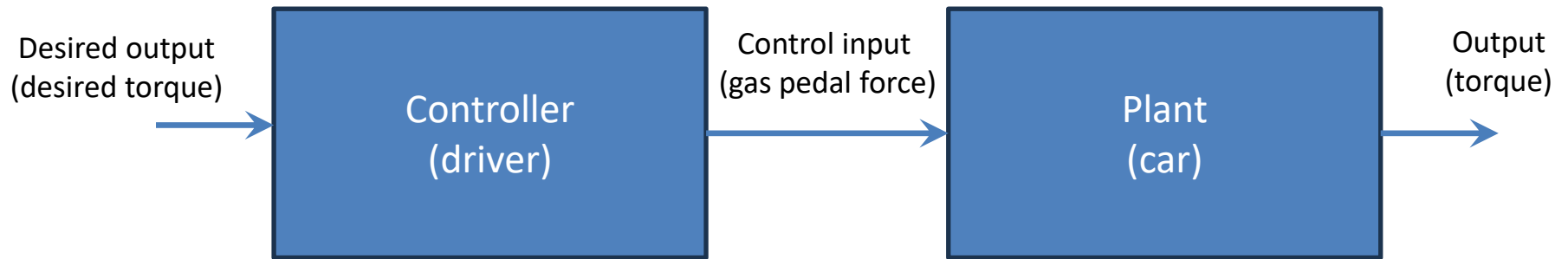


Basics of Feedback Theory

Basics of Feedback Theory



An open-loop control system: the controller applies the control input without knowing the plant output

Basics of Feedback Theory

$$\text{Transfer Function} = \text{Output} / \text{Input}$$

Example:

A potentiometer is used as a position sensor. The pot is configured in such a way that 0° of rotation yields 0 V and 300° yields 10 V. Find the transfer function of the pot.

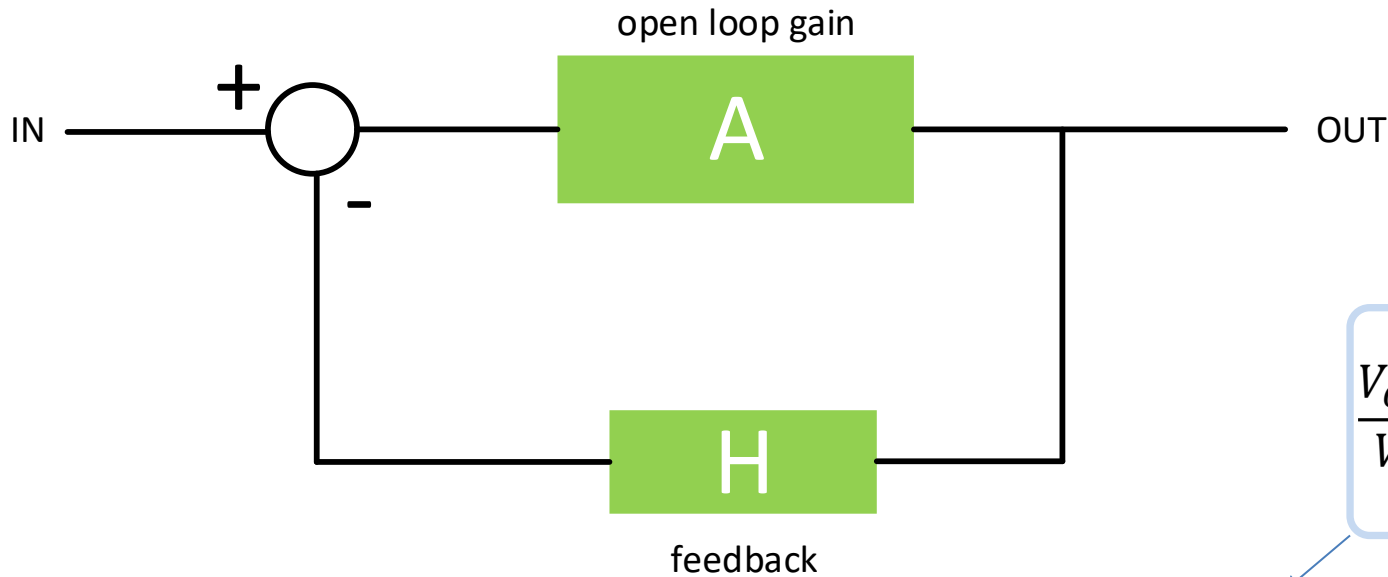
SOLUTION

The transfer function is output divided by input. In this case, the input to the pot is “position in degrees,” and output is volts:

$$\text{TF} = \text{output/input} = 10 \text{ V} / 300^\circ = 0.0333 \text{ V/degrees}$$

The transfer function of a component is an extremely useful number. It allows you to calculate the output of a component if you know the input.

Basics of Feedback Theory



$$\frac{V_{OUT}}{V_{IN}} = \frac{A}{1 + HA}$$

Diagram illustrating the closed loop gain equation. The equation is shown with annotations: $\frac{V_{OUT}}{V_{IN}}$ is labeled "closed loop gain", A is labeled "open loop gain", and HA is labeled "loop gain".

$$V_{OUT} = A (V_{IN} - HV_{OUT})$$

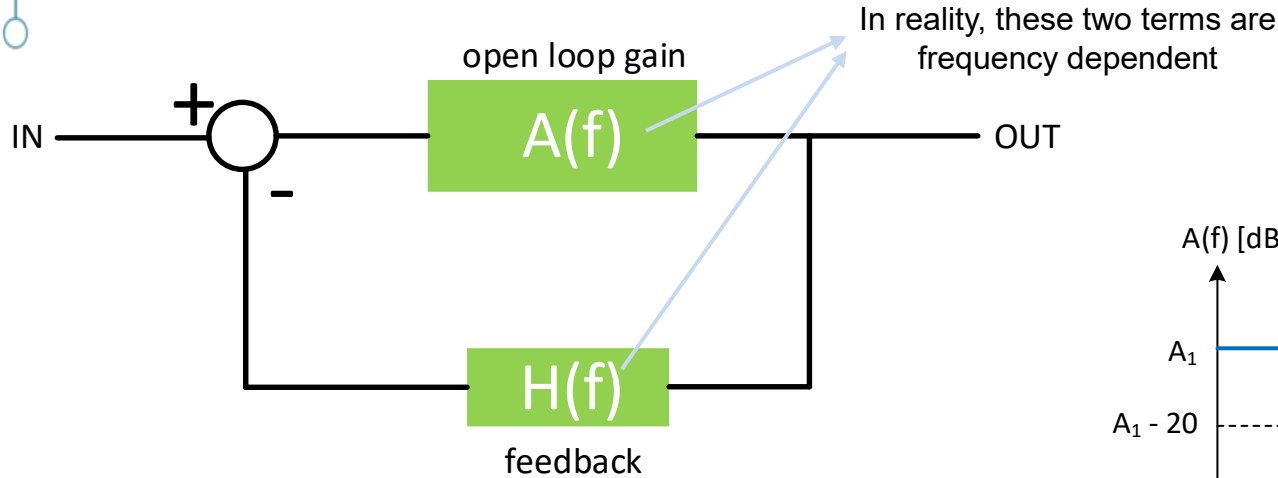
$$\frac{V_{OUT}}{V_{IN}} = \frac{A}{1 + HA} \quad \text{If } A \rightarrow \infty$$

$$\frac{V_{OUT}}{V_{IN}} = \frac{1}{H}$$

closed loop gain

In reality, open loop gain is not stable due to temperature, non-linearity etc. With negative feedback, the closed loop gain can be stabilized since the gain depends only on the negative feedback factor

Basics of Feedback Theory



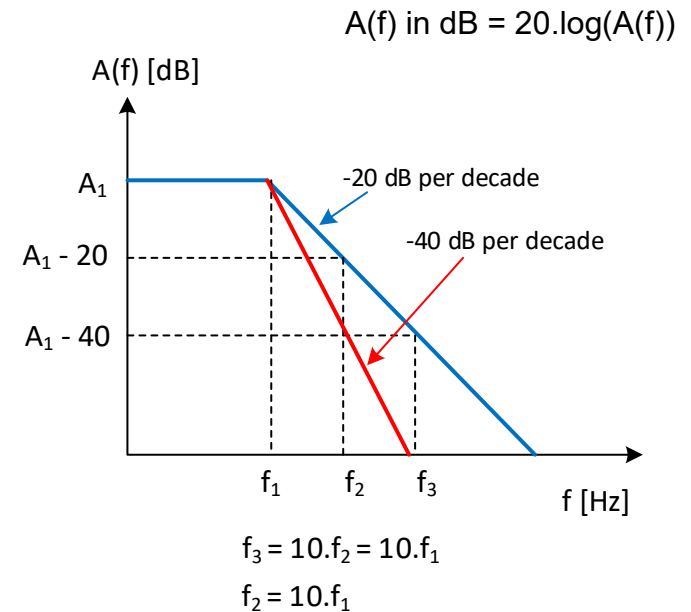
$$V_{OUT}(f) = A(f) (V_{IN}(f) - H(f)V_{OUT}(f))$$

$$\frac{V_{OUT}(f)}{V_{IN}(f)} = \frac{A(f)}{1 + H(f)A(f)}$$

closed loop gain

open loop gain

loop gain



Please note that $V_{OUT}(f)/V_{IN}(f) \rightarrow \infty$, if $H(f)A(f) = -1$ at a certain frequency

In other words, $V_{OUT}(f)/V_{IN}(f)$ **becomes unstable** → Negative feedback must be implemented carefully!

Whether you have a 1. order or 2. order system depends on the number of the poles

Blue: 1.order low-pass characteristics

Red: 2. order low-pass characteristics

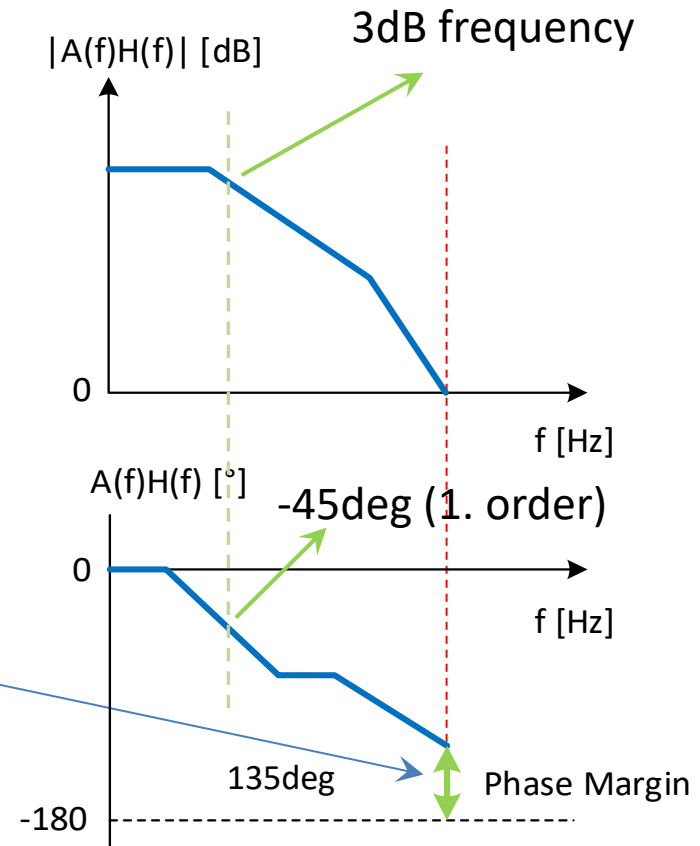
Basics of Feedback Theory

closed loop gain $\leftarrow \frac{V_{OUT}(f)}{V_{IN}(f)} = \frac{A(f)}{1 + H(f)A(f)} \rightarrow$ open loop gain

loop gain

Please note that $V_{OUT}(f)/V_{IN}(f) \rightarrow \infty$, if $H(f)A(f) = -1$ or in other words, loop gain at a certain frequency

A larger phase margin implies a more stable system



Literature

- Advances in Industrial Control, Practical PID Control, Antonio Visioli, Springer Verlag, 2006
- MATLAB Central - Mathworks
- Modern Control Design with MATLAB and Simulink, Ashish Tewari, John Wiley & Sons Ltd, 2002